

UNIT I (Basics of Design & Welding)

What are the various phases in design process?

- a. Recognition of need
- b. Definition of problem
- c. Synthesis
- d. Analysis and Optimization
- e. Evaluation
- f. presentation

How the machine design may be classified?

- a. Adaptive design
- b. Developed design
- c. New design
- d. Rational design
- e. Empirical design
- f. Industrial design

What are the types of loads that can act on machine components?

- a. Steady load
- b. Variable load
- c. Shock load
- d. Impact load

Differentiate between resilience and toughness.

- ✓ Resilience is the property of the material to absorb energy and to resist shock and impact loads.
- ✓ Toughness is the property of the material to resist fracture due to high impact load.

Define Creep.

When a part is subjected to a constant stress at high temperature for a long period of time, it will undergo a slow and permanent deformation called creep.

What are the factors affecting selection of material for machine element

- 1. Load applied
- 2. Purpose and operating conditions of the part.
- 3. Suitability for manufacture.
- 4. Minimum weight and optimal size
- 5. Availability and cost.

What is optimum design?

In iterative nature of design, we can arrive at a design which provides the best possible technical conditions like maximum on the best performance of the product with the least cost per unit, consistent with weight, size, strength and similar considerations. Such a design is called Optimum design.

Define factor of safety.

Factor of safety (FOS) is defined as the ratio between the ultimate stress and working stress.

List out the factors involved in arriving at factor of safety.

- ✓ Material Properties
- ✓ Nature Of Load
- ✓ Mode Of Failure

What are the types of variable stresses?

- a. completely reversed or cyclic stresses
- b. fluctuating stresses
- c. repeated stresses
- d. alternating stresses

Define principal stress.

The direct stresses acting along the principal planes (which have no shear stress) in a strained material is known as principal stresses.

Differentiate between repeated stress and reversed stress.

Repeated stress refers to a stress varying from zero to a maximum value of same nature. Reversed stress or cyclic stress varies from one value of tension to the same value of compression.

Define factor of safety for fatigue loading.

Factor of safety for fatigue loading = endurance limit stress/Design stress

Define Endurance limit.

Endurance limit is defined as the fatigue stress below which the specimen will not fail for infinite number of cycles (10^7 cycles).

State different theories of failures.

1. Maximum principal stress theory (or) Rankines theory
2. Maximum shear stress theory (or) Guest's theory
3. Maximum principal strain theory (or) Saint Venant theory
4. Maximum distortion energy theory
5. Maximum strain energy theory

State St.Venant theory of failure.

According to this theory, failure occurs when the maximum strain developed in the machine member is equal to maximum strain at yield point in a tension test.

What are the appropriate theories of failure for ductile and brittle materials?

For ductile materials – maximum distortion energy theory

For brittle materials – maximum principle stress theory

State maximum shear stress theory.

According to maximum shear stress theory failure occurs when the maximum shear stress developed in a machine member is equal to the maximum shear stress at yield point in a tension test.

Explain reasons for using different theories of failures.

The problem of predicting the failure stresses for members subjected to bi axial or tri axial stresses is much complicated. Hence we are using different theories of failure.

Define the term fatigue.

When the material is subjected to repeated stresses, it fails at stresses below the yield point stresses, such type of failure is known as fatigue.

Define the morphology of design.

Morphology of design consisting of problem formulation analysis search for alternative an evaluation decision taking and specification of the solution.

Stress Intensity Factor?

Stress Intensity factor = Maximum stress/Nominal Stress

Define fit and tolerance?

Fit is the relation between the two mating parts in which one is inserted into the other with degree of tightness or looseness.

Tolerance is the maximum permissible value within which the part can vary.

Write down the factors influencing machine design?

- ✓ Cost
- ✓ Efficiency
- ✓ Strength
- ✓ Stiffness
- ✓ Wear Resistance
- ✓ Dimension

What are the main indications of complete weld symbol?

- | | |
|----------------------|-------------------------|
| 1. Reference line | 2. Arrow |
| 3. Basic weld symbol | 4. Dimensions |
| 5. Tail | 6. Supplementary symbol |
| 7. Finish symbol | 8. Specification |
| 9. Process | |

What are the main types of welding?

- | | | |
|------------------|-------------------|--------------------------------|
| 1. Forge welding | 2. Fusion welding | 3. Electric resistance welding |
|------------------|-------------------|--------------------------------|

What are various types of welded joints?

- | | | |
|--------------------------|--------------------|--------------------|
| 1. Lap (or) fillet joint | | |
| a. Transverse fillet | b. Parallel fillet | c. Circular fillet |
| 2. Butt joint | | |
| a. Square butt | b. V butt | c. U butt |
| 3. Corner joint | | |
| 4. Edge joint | | |
| 5. T joint | | |

What are the advantages of welding?

1. Welded joints are higher in weight and have higher efficiency
2. Welded joints are leak proof
3. Economical from the point of cost of material and labours.
4. The design can be easily
5. Less time for production.

State the limitation of welding.

1. It has poor vibration damping characteristics.
2. Welding results distortion of parts which induces residual stresses.

What are eccentrically loaded welded joints?

If the external load applied on the welded joint is not passing through its geometric centre then it is called as eccentrically loaded welded joint.

What are stresses induced in eccentrically loaded welded joint?

1. Direct shear stress
2. Bending (or) torsional shear stress

What are the types of eccentrically loaded welded joints?

1. Welded joint subjected to moment in the plane of the weld.
2. Welded joint subjected to moment in a plane normal to the plane of weld.

Why are welded joints preferred over riveted joints?

Riveted Joints	Welded Joints
1. Metal plates are to be drilled and joined by rivets.	1. Drilling work is eliminated, plates can be directly welded.
2. It may require covering plates	2. No covering plate is required.

What are uncertainties to be considered in design of welds?

1. Stress concentration
2. Metallurgical change
3. Residual stresses

What are the assumptions made in design of welded joint?

- ✓ It is assumed that the tensile stress is distributed uniformly across the section of the butt weld.
- ✓ It is assumed that the shear stress in a parallel fillet weld is uniformly distributed along the entire length of the weld.
- ✓ Stress situation in a fillet weld is complicated because of bending action of the force.

Unit II (Screw Thread and Power Screw)**What is a power screw?**

Power screw is a machine component consists of a nut and a long bolt, used to convert rotary motion into translatory motion, hence called as translation screw also.

Where is the power screws employed?

Power screws are employed in vices, universal testing machine, presses, screw jack, lathe, sluice gate of dam, C-clamps etc.

Specify the types of threads used for power screws.

- Square thread.
- Trapezoidal thread or Acme thread.
- Buttress thread.

Name the materials for power screw components.

Usually the screw-rods are made of carbon steels or alloy steels. The nuts are made of various grades of cast-iron or bronze in order to reduce friction between screw and nut.

Narrate the relationship between lead.

Lead = Pitch x number of starts

For example, in double start thread, the lead is equal to two times the value of pitch.

What is a self-locking screw? (or) In what condition, a power screw will become a self-locking screw

If the lifted load by some effort through power screw is automatically lowered without any effort, then that type of screw is called as overhauling screw. If any effort is required to lower the load, that screw is known as self-locking screw.

What are the various stresses induced in power screws?

- Tensile and compressive stresses due to axial load.
- Torsional shear stress.
- Shear stress due to axial load.
- Bearing stress.

What is Screw thread?

A Helical groove made on the peripheral of the cylindrical rod is called screw thread.

What are the types of threaded fasteners?

Screws, bolts, nuts and studs etc.

What kinds of stresses are induced in threaded fasteners?

- Axial tensile stress and
- Torsional shear stress.

Differentiate screw, bolt, stud and nut in their structures

- ✓ In screw or bolt a single helical groove is made from one end to other end, but the screw is tapered from one end to other end, whereas the bolt is having constant diameter throughout its length.
- ✓ In stud, helical grooves are made in two different directions, leaving the centre portion unscrewed. Screws and Bolts are having heads, but studs are not having heads

By what materials threaded fasteners are made?

Steel is the material of which most of the fasteners are made. For improving their properties alloy steels like nickel-steel, nickel-chromium steel are preferred.

Define the following terms.

(a) Pitch

(b) Lead

Pitch: It is defined as the axial distance between the adjacent threads.

Lead: It is nothing but the distance between two corresponding points in the same helix. In the multi-start thread, lead is equal to the product of pitch and the number of starts.

How are screw-threads classified?

Screw threads are classified based on 1. Usage as

- Fastening threads
- Power threads

How is a screw thread designated?

In India metric threads are adopted as standard threads. There are two types of metric threads namely **coarse and fine threads**. Coarse thread is designated by the capital letter M followed by the major diameter of the thread in mm. For example: M10. The pitch is additionally indicated for fine thread. For example: M10 x1.25.

In what way coarse thread is differed from fine thread?

Fine and coarse threads are having same major and minor diameters except their pitch values. Fine threads are having smaller pitches than coarse threads.

What is meant by single start and multi-start threads?

If the thread is having only one helical groove, then it is called single-start thread, whereas the thread having more than one helical groove, (upto four grooves) is called as multi-start threads. If the number of starts is two, then it is known as double start thread and so on.

What are the merits and demerits of screw joints?**Merits:**

- Screw joints are highly reliable in operation.
- Simple manufacture with the possibility of maintaining high accuracy.
- They can be made in any convenient shape and small size. In addition to fastening, screws and nuts are used as power screws.

Demerits:

- A stress concentration is available in threaded portions and hence lowering of their life.
- Self-loosing properties and hence air-tight joints can not be maintained, unless providing some locking devices.

What are the various methods of preventing thread loosening?

Thread loosening can be prevented by

- Providing locking devices such as lock nut, castle nut etc.
- Selecting the screw having large number threads per unit length etc.

A bolt is designated as M24 x 2. What these two numbers signify?

In the bolt designation, M24 x 2 means that the bolt is having 24 mm nominal diameter and 2 mm pitch length.

Unit III (Key, Coupling, Clutch and Brake)**What is key?**

Key is an element which is used to connect two machine parts for preventing relative motion of rotation with respect to each other.

Name the types of keys.

- | | |
|---------------|----------------------------|
| 1. Saddle key | 2. Tangent key |
| 3. Sunk key | 4. Round key and taper pin |

How sunk keys are provided?

Sunk keys are provided half in the key way of the shaft and half in the key way of the hub or boss of the pulley.

List various types of sunk keys.

- | | |
|-------------------------|--------------------|
| 1. Rectangular sunk key | 2. Square sunk key |
| 3. Parallel head key | 4. Gib head key |
| 5. Feather key | 6. woodruff key |

What is a keyway?

Keyway is a slot or recess in a shaft and hob of the pulley to accommodate a key.

What is Gib head key? What are the advantages?

In a rectangular sunk key with a head at one end is known as gib head key. It is usually provided to facilitate the removal of key.

What is woodruff key? State its application.

It is piece from a cylindrical disc having segmental cross section. A woodruff key is capable of tilting in a recess milled out in the shaft by a cutter having the same curvature as the disc from which the key is made.

List the various purposes of shaft couplings?

1. To provide for the connection of shafts of units that is manufactured separately and to provide for disconnection for repairs or alternations.
2. To provide misalignment of the shafts or to introduce mechanical flexibility.
3. To introduce protection against overloads.
4. To reduce the transmission of shock loads from one shaft to another.

List out the requirements of a shaft coupling?

1. It should be easy to connect or disconnect.
2. It should transmit the full power of the shaft
3. It should hold the shafts in perfect alignment.

What is rigid coupling? What are its types?

It is used to connect two shafts which are perfectly aligned. The types are

1. Sleeve or muff coupling
2. Clamp or split muff or compression coupling
3. Flange coupling.

What is flexible coupling? What are its types?

Flexible coupling is a type of coupling used to connect two shafts having both lateral and angular misalignment.

Types: a) Bushed pin type coupling

b) Universal coupling

c) Oldham's coupling

What is a flange coupling?

It is a coupling having two separate cast iron flanges. Each flange is mounted on the shaft end and keyed to it. One of the flanges has a projected portion and the other flange has a corresponding recess. This helps to bring the shafts into line and maintain alignment.

What are various types of flange coupling?

1. Unprotected type flange coupling
2. Protected type flange coupling
3. Marine type flange coupling

What is the difference between rigid and flexible coupling?

Rigid coupling is used to connect two shafts which are perfectly aligned.

Flexible coupling is used to connect two shafts having both lateral and angular misalignment.

List any two methods used for manufacturing of shafts.

1. Cold rolling
2. Hot rolling
3. Turning or grinding from rough bars.

What is the effect of keyway cut into the shaft?

The keyway cut into the shaft reduces the load carrying capacity of the shaft. This is due to the stress concentration near the corners of the keyway and reduction in the cross sectional area of the shaft.

What is the difference between coupling and a clutch?

A coupling is a device used to make permanent or semi-permanent connection where as a clutch permits rapid connection or disconnection at will of the operator.

What is the mode of failure of the bolts in a flange coupling?

Direct shear stress failure due to torque transmission.

How couplings are specified?

- a) Diameter of shaft
- b) Diameter of sleeve or muff
- c) Length of sleeve or muff
- d) Outer diameter of hub
- e) Nominal diameter or bolt
- f) PCD of bolt circle.

What is clutch?

A clutch is a machine member used to connect a driving shaft to a driven shaft so that the driven shaft may be started or stopped at will, without stopping the driving shaft.

What are the types of clutch?

Two types of clutches

- ✓ Positive clutches
 - Jaw or claw clutch.
- ✓ Friction clutches
 - Disc or plate clutch
 - Cone clutch
 - Centrifugal clutch.

What is meant by positive clutch?

A clutch which transmits power from driving shaft to the driven shaft by means of jaw or teeth is called positive clutch.

In what way the clutches are differed from brake?

A clutch is used to engage the driving and driven members temporarily and keep them moving together, where as brakes are employed to stop a moving member or to control its speed.

What is brake?

A brake is a device by means of which artificial frictional resistance is applied to a moving machine member, in order to retard or stop the motion of a machine.

What are the types of brake?

The brakes, according to the means used for transforming the energy by the braking element, are classified as :

- ✓ Hydraulic brakes *e.g.* pumps or hydrodynamic brake and fluid agitator,
- ✓ Electric brakes *e.g.* generators and eddy current brakes, and
- ✓ Mechanical brakes.

What are self-energizing brakes and self-locking brake?

The frictional force helps to apply the brake. Such type of brakes are said to be **self-energizing brakes**. When the frictional force is great enough to apply the brake with no external force, then the brake is said to be **self-locking brake**.

Unit IV (spring, cotter & knuckle joint)

What is spring and where it is employed?

A spring is an elastic body, which distorts when loaded and recover its original shape when the load is removed. It finds applications in many places such as automobiles, railway wagons, brakes, clutches, watches and so on.

By what materials springs can made?

Springs are made of oil tempered carbon steel containing 0.6% to 0.7% carbon and 0.6% to 1% manganese. Phosper bronze, beryllium, copper are used for special purpose.

What are functions of the spring?

- ✓ To measure forces in spring balance, meters and engine indicators.
- ✓ To store energy.

Name various types of springs.

Helical springs, Spiral springs, leaf springs and disc (or) Belleville spring.

What is Spring Index (C)?

It is the ratio of mean pitch diameter to the diameter of the wire. ($C = D/d$)

What are Active and Inactive coils?

The coils which are free to deflect under load are called active coils and the coils which do not take part in deflection of a spring are called inactive coils.

Define the term “Spring Rate” or stiffness (q)?

It is defined as the load required per unit deflection. It is also called as stiffness of the spring.

Define surging of springs

The spring material is subjected to higher stresses, which may cause early fatigue failure of springs. This effect is called as surging of springs.

What material is used for leaf spring?

Plain carbon steel having 0.9% to 1% carbon in annealed condition is normally used for leaf springs. Chrome vanadium and silica manganese steels are used for the better grade springs.

What is nipping of laminated leaf spring?

Nip is the initial gap which is provided in between the full length leaves and graduated leaves of the leaf spring.

What are the end conditions of springs?

- a. Plain en.
- b. Plain and ground end.
- c. Squared end.
- d. Squared and ground end.

What is buckling of springs?

The helical compression springs behaves like a column and buckler at a comparative small load when the length of the spring is more than four times the mean coil diameter.

Why Wahl's factor is to be considered in the design of helical compression spring?

When wire wound in the form of helix, compressive stress is induced in the inner side of spring and tensile stress is induced in outer side of spring. Due to this stress concentration is produced in outer side of spring. So Wahl's factor is to be considered in the design of helical compression spring.

Define solid length of helical spring.

When compressions spring is compressed until the coils come in contact with each other, then the spring is said to be solid and resulting length is called solid length.

Define free length of a helical spring?

It is the length of the spring in free or unloaded condition.

Define pitch of the spring coil.

Pitch of the coil is defined as the axial distance between adjacent coils uncompressed state.

What is torsion springs?

Torsion springs may be of helical or spiral type. The helical type may be used only in applications where the load tends to wind up the spring and are used in various electrical mechanisms. The spring type is also used where the load tends to increase the number of coils and when made of flat strip are used in watches and clocks.

How the stiffness of a spring can be increased?

The stiffness of a spring can be increased by decreasing the number of turns.

Write the advantage of leaf spring over helical springs?

Leaf springs are made out of flat plates. The advantage of leaf spring over helical spring is that the end of the spring may be guided along a definite path as it deflects to act as a structural member in addition to energy absorbing device.

What is Cotter Joint? Application

- ✓ A cotter is a flat wedge shaped piece of rectangular cross-section and its width is tapered from one end to another for an easy adjustment.
- ✓ A cotter joint is a temporary fastening and is used to connect rigidly two co-axial rods or bars which are subjected to axial tensile or compressive forces.
- ✓ It is usually **used in** connecting a piston rod to the crosshead of a reciprocating steam engine, a piston rod and its extension as a tail or pump rod, strap end of connecting rod etc.

What are the types of Cotter Joints?

Following are the three commonly used cotter joints to connect two rods by a cotter:

- ✓ Socket and spigot cotter joint,

- ✓ Sleeve and cotter joint, and
- ✓ Gib and cotter joint.

What is Knuckle Joint? Application.

- ✓ A knuckle joint is used to connect two rods which are under the action of tensile loads.
- ✓ A knuckle joint may be readily disconnected for adjustments or repairs.

Its use may be found in the link of a cycle chain, tie rod joint for roof truss, valve rod joint with eccentric rod, pump rod joint, tension link in bridge structure and lever and rod connections of various types.

UNIT-V (SHAFT)

Define shaft.

A shaft is a rotating machine element which is used to transmit power from one place to another. Shaft is used for the transmission of torque and bending moment.

Differentiate between shaft and axle.

An axle, though similar in shape to the shaft, is a stationary machine element and is used for transmission of bending moment only. It simply acts as a support for some rotating body.

What is spindle?

A spindle is a short shaft that imparts motion either to a cutting tool or to a work piece.

What are the materials used for shafts.

For ordinary shafts – mild steel

For high strength shafts – alloy steel such as Nickel, Ni-Cr steels (or) Cr – V steels.

What are the types of shafts and their importance?

- ✓ Transmission shafts – These shafts transmit power between the source and the machines absorbing power. These shafts carry machine parts such as pulleys, gears etc
- ✓ Machine shafts – these shafts form an integrated part of the machine itself. The crankshaft is an example of machine shaft.

What are various types of stresses induced in the shafts?

- ✓ Shear stresses due to transmission of torque.
- ✓ Bending stresses.
- ✓ Stresses due to combined torsional and bending loads.

On what basis the shafts are designed.

- ✓ Based on rigidity and stiffness
- ✓ Based on strength
- ✓ Based on critical speed.

Differentiate the hollow shaft and solid shaft.

The hollow shafts are used in marine work. These shafts are stronger per kg of material and they may be forged on a mandrel, thus making the material more homogenous than a solid shaft.

What are the desirable properties for the materials for shafts and axles?

- ✓ Sufficient high strength
- ✓ A low sensitivity to stress concentration
- ✓ Ability to withstand heat and case hardening treatment.
- ✓ Good machinability

Why rotating shaft are generally made with circular cross section?

Stress distribution pattern will be uniform throughout the circular cross section.

Define Torsional stiffness of shaft.

It is defined as the resisting strength of a shaft to torsional load. Mathematically it can be calculated by the formula.

Define critical speed of a shaft.

Rotating shaft tends to vibrate violently in transverse direction at certain speeds known as critical (or) whirling speed.

State any two reasons for preferring hollow shaft over solid shaft.

- ✓ For same weight of shaft, hollow shaft can transmit 1.5 times the torque transmitted by solid shaft.
- ✓ For a particular power transmission hollow shaft requires minimum weight.

What theories of failure are applied in shaft design?

- ✓ Maximum principal stress theory (or) Rankine's theory
- ✓ Maximum shear stress theory (or) Guest's theory

State any two methods of solving problems involving combined steady or variable stresses.

- a) Gerber method b) Goodman method c) Soderberg method

Define Goodman line.

It is a straight line connecting the endurance limit and ultimate strength in variable stress diagram and Goodman line is applicable for designing of brittle materials.

Define Soderberg line.

- ✓ Gerber curve is a parabola drawn between endurance limit and ultimate tensile strength.
- ✓ Soderberg line is a straight line connecting endurance limit and the yield strength in variable stress diagram. Soderberg line applicable for designing of ductile materials.

What is stress concentration?

- ✓ The irregularity in stress distribution caused by abrupt changes of form is called stress concentration.

What is stress concentration factor?

- ✓ Stress concentration factor = maximum stress at the change of cross section / nominal stress.

What are different methods to reduce stress concentration?

- ✓ Providing Fillets
- ✓ Drilling Additional Holes
- ✓ Providing Additional Grooves
- ✓ Drilling Additional Counter Sunk.

What is notch sensitivity factor?

It is defined as the degree to which the theoretical effect of stress concentration is actually reached.